RALLY POINT

Multi-Tenant SaaS & Superadmin Architecture

Engineering Handoff & Technical Release Specification

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Upstream PR: https://github.com/zylush/rally-point-web/pull/3	**Date of Release:** October 14, 2026
Status: Fully Verified (100% Test Pass Rate)	**Stack:** React 19, TypeScript 5.8, Vite 8, Supabase RLS

1. Executive Overview & Architecture Vision

Rally Point has evolved from a single-facility court rental management web application into a full-scale **Multi-Tenant Software-as-a-Service (SaaS) Platform**. Under this new architectural paradigm, multiple autonomous pickleball facilities operate as independent branches (e.g., General Santos City, Davao City, Cebu IT Park), while an executive **Platform Superadmin** oversees all branches, manages subscriptions/status, and automatically collects a configurable percentage commission on every confirmed court booking.

> **Key Architectural Achievement:** Complete multi-tenant data isolation and revenue attribution are maintained both in offline demo mode and production Supabase SQL. Every transaction calculates:
> **Commission Cut:** Booking Amount × (Branch Commission Rate / 100) | **Branch Net:** Booking Amount - Commission Cut

2. Scope of Deliverables & Component Breakdown

Module	Key Files Modified / Added	Description & Business Capability
Domain Models & Types	`src/types.ts`	Added 'superadmin' role, Branch, PlatformCommission, and SuperadminStats interfaces. Bound branch_id across members, courts, bookings, and inquiries.
Multi-Tenant Store Engine	`src/lib/demoStore.ts` `src/lib/api.ts`	Multi-branch seeding (Gensan, Davao), superadmin authentication, automatic commission ledger creation on payment confirmation, and branch CRUD.
Superadmin Portal	`src/pages/Superadmin.tsx`	Executive dashboard featuring 4 global KPI cards, interactive Branch Directory with suspend/activate controls, Provision Branch modal, and real-time Commission Ledger.
Navigation & Shell	`src/components/Shell.tsx` `src/App.tsx`	Configured Superadmin desktop/mobile 6-tab navigation, protected /superadmin routing, and cross-branch administrative inspection access.
Quick Demo Auth	`src/pages/LoginPage.tsx`	Added 1-click Superadmin login button in 4-column responsive grid alongside Admin, Staff, and Member credentials.
Database Migration	`supabase/migrations/20261014000000_*.sql`	PostgreSQL migration schema creating branches and platform_commissions tables, foreign keys, private.current_branch_id() helper, and RLS tenant policies.

Codebase Polish & Hygiene	`src/pages/admin.tsx`
	`src/pages/HelpPage.tsx`
	`src/pages/MemberPass.tsx`
		Purged 360+ lines of dead shader code, eliminated dual-header rendering in admin views, implemented offline SVG QR generation, and added inquiry status filter tabs.

3. Multi-Tenant Commission & Revenue Architecture

The monetization model is built directly into booking payment confirmation flows. Branch managers operate their club facilities autonomously, but the platform governance tier enforces a contractual fee split:

- **Step 1 (Booking Creation):** Member or Staff initiates a court reservation for an hourly fee at a specific branch.
- **Step 2 (Payment Confirmation):** Payment is confirmed via GCash, Maya, Cash, or Card through confirmBookingPayment().
- **Step 3 (Commission Calculation):** The engine pulls the branch's specific commission_rate (default 5.0%).
- **Step 4 (Ledger Settlement):** An immutable PlatformCommission entry is recorded with gross booking amount, platform commission cut, and branch net amount.
- **Step 5 (Global Aggregation):** The Superadmin KPI overview dynamically recalculates platform lifetime gross volume and total commission revenue.

Branch Name	Hourly Rate	Duration	Gross Booking	Platform Cut	Superadmin Earns	Branch Net
Rally Point Gensan	Php 500.00	2 Hours	Php 1,000.00	5.0%	**Php 50.00**	Php 950.00
Rally Point Davao	Php 650.00	2 Hours	Php 1,300.00	6.0%	**Php 78.00**	Php 1,222.00
Cebu IT Park Hub (New)	Php 700.00	3 Hours	Php 2,100.00	7.5%	**Php 157.50**	Php 1,942.50

4. Quality Assurance & Verification Matrix

The codebase has been verified against three independent verification gates with strict zero-regression standards:

Verification Gate	Command Executed	Scope Checked	Result
Unit & Integration Tests	`npm.cmd test -- --run`	14 test suites, 51 test cases covering auth security, superadmin KPIs, commission calculations, branch provisioning, and UI tabs.	**51/51 PASSED (100%)**
Static Linter & Code Hygiene	`npm.cmd run lint`	oxlint analyzing 43 files against 104 rules (detecting unused imports, dead references, and variable leaks).	**0 Warnings, 0 Errors**
TypeScript & Production Build	`npm.cmd run build`	Strict tsc -b type checking and Vite production bundling with asset compression.	**Clean Build (1.30s)**

5. Interactive Verification & Demo Accounts

Reviewers can test all workflows locally at **http://localhost:5173/rally-point-web/**. The login screen features 1-click quick demo buttons for all 4 roles:

Role	Email	Password	Primary View & Responsibilities

Superadmin	`superadmin@rallypoint.local`	`superadmin123`	Oversees all branch sites, provisions new branches, views global KPIs, and audits commission cuts.
Admin	`admin@rallypoint.local`	`admin123`	Branch manager for Gensan — oversees courts, schedule board, bookings, users, and staff inquiries.
Staff	`staff@rallypoint.local`	`staff123`	Front desk operations — checks in members via QR, manages active court timers, and submits help inquiries.
Member	`member@rallypoint.local`	`member123`	Player experience — reserves courts, signs up for open play sessions, and views personal QR pass.

6. Production Deployment & Handoff Checklist

When transferring from local/demo mode into hosted Supabase staging and production environments:

- ■ **Apply Database Migration:** Execute supabase/migrations/20261014000000_multi_tenant_superadmin.sql to create branches, platform_commissions, and RLS policies.
- ■ **Configure Tenant Domain Routing:** Configure DNS wildcard routing (e.g. *.rallypoint.ph) or URL path matching to bind user sessions to their designated branch_id.
- ■ **Automated Settlement Payouts:** In future iterations, link platform_commissions to a payment provider API (e.g. PayMongo Split Payouts or Stripe Connect) for automatic direct deposit of branch net revenue.
- ■ **Staff/Admin Provisioning Policy:** Enforce that branch Admin and Staff profiles are only created by the Superadmin or Club Manager rather than the public join form.

Handoff Status: COMPLETED & COMMITTED	**Git Remote / Branch:** `bapdagreat/rally-point-tenancy-handoff`
Pull Request: https://github.com/zylush/rally-point-web/pull/3	**Release Engineer:** Dayle Angelo Peña (@DayleeQuest)